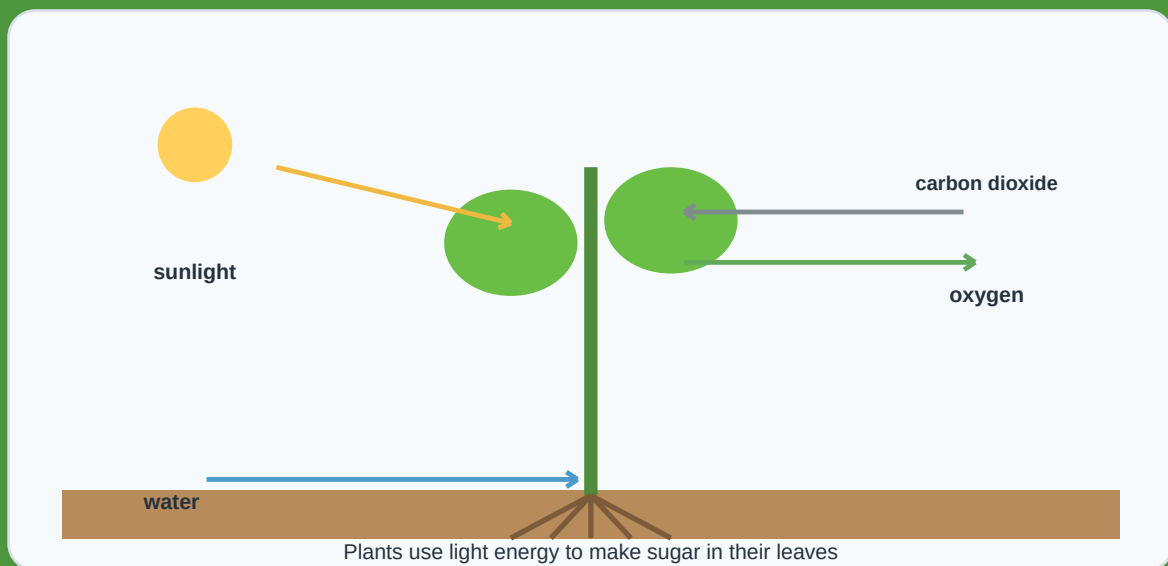


KATHAQUEST TEST CHAPTER

Photosynthesis: How Plants Make Food

Discover how leaves use sunlight, water and carbon dioxide



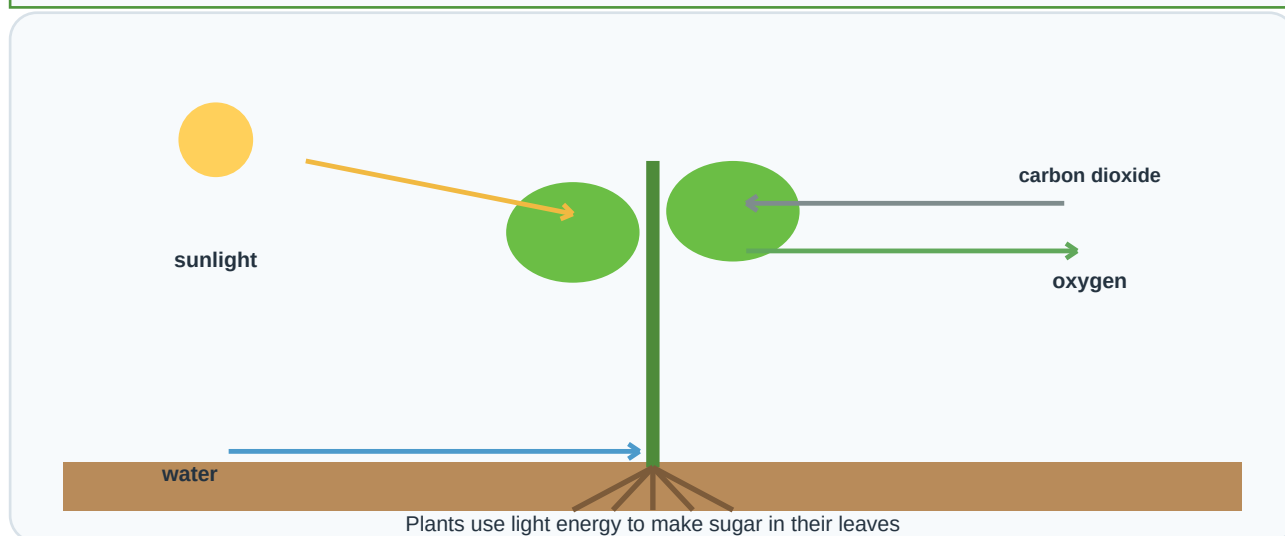
Recommended reading level: Ages 8-12

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What you will learn

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|---|---|
| 1 | Explain what photosynthesis is. |
| 2 | Identify the inputs and products of photosynthesis. |
| 3 | Describe how roots, stems and leaves work together. |
| 4 | Connect plant growth with food chains and oxygen in the atmosphere. |

Key words: photosynthesis, chlorophyll, carbon dioxide, oxygen, glucose, stomata, roots



1. Plants are food makers

Animals must eat other organisms for energy, but green plants can make their own food. They do this through **photosynthesis**, a process that captures light energy and stores it in sugar.

Most photosynthesis happens in leaves. Leaves are broad and thin, giving them a large surface for collecting light and a short distance for gases to travel.

The word photosynthesis can be split into photo, meaning light, and synthesis, meaning putting together.

2. The ingredients

Roots absorb water and minerals from the soil. Water travels upward through narrow tubes in the stem. Carbon dioxide enters the leaf through tiny openings called **stomata**.

Inside leaf cells are structures called chloroplasts. They contain **chlorophyll**, the green pigment that absorbs light energy.

Visual moment to look for: water moving from roots to leaves and carbon dioxide entering through stomata.

3. Turning light into stored energy

Using light energy, plants rearrange water and carbon dioxide to make **glucose**, a simple sugar. Oxygen is released as a product.

A simplified word equation is: carbon dioxide + water + light energy → glucose + oxygen. The actual process has many steps, but this equation shows the main inputs and outputs.

Plants do not get their food from soil. Soil provides water and minerals, while the plant builds sugar mainly from carbon dioxide and water.

4. What plants do with glucose

Plants use glucose during respiration to release energy for living processes. They can join glucose molecules to make starch for storage or cellulose for strong cell walls.

They also use sugar, together with minerals from the soil, to build proteins, oils, flowers, fruits and new leaves. This is how a small seedling grows into a large plant.

Visual moment to look for: a time-lapse of a seed germinating and growing toward light.

5. Why photosynthesis matters

Photosynthesis supports almost every food chain. Herbivores eat plants, and predators may eat herbivores. Even people who eat meat depend indirectly on plants.

The process also releases oxygen used by many organisms during respiration. Forests, grasslands, crops and microscopic algae all contribute to the movement of carbon and oxygen through ecosystems.

Plants also respire. They make sugar through photosynthesis and release usable energy from sugar through respiration.

Chapter summary

- Photosynthesis uses light energy to make glucose from carbon dioxide and water.
- Roots supply water, while stomata allow gases to enter and leave leaves.
- Chlorophyll absorbs the light needed for photosynthesis.
- Plants use glucose for energy, storage and growth.
- Photosynthesis supports food chains and releases oxygen.

Glossary

Word	Meaning
photosynthesis	The process by which plants use light to make glucose.
chlorophyll	The green pigment that absorbs light energy.
stomata	Tiny openings that control gas exchange in leaves.
glucose	A simple sugar that stores chemical energy.
carbon dioxide	A gas used by plants during photosynthesis.
oxygen	A gas released during photosynthesis and used in respiration.

Quick check

1. What are the three main inputs needed for photosynthesis?
2. Where does most photosynthesis happen?
3. How does water reach the leaves?
4. What can a plant make from glucose?
5. Why is photosynthesis important to animals?

Testing note: This chapter is intentionally written with clear concepts, visual events and child-friendly language. It is suitable for testing chapter extraction, concept detection, media search, narration, quiz generation and revision-video workflows.